Lecture 9

Implicit derivatives and mean value theorems

Uses.

Summary 5: Continuity is relative to the domain. Order passes to finite limits, including one-sided limits.
Summary 6: The intermediate value theorem supplies roots between opposite endpoint signs. The constructed constant e satisfies $2 < e < 3$.
Summary 7: Continuous functions on closed bounded intervals attain their extreme values. A derivative is a finite quotient limit and implies continuity.
Summary 8: Arithmetic, chain and inverse rules apply with their stated domains. Elementary derivative formulas are proved, and a derivative need not be continuous.

Objectives. *A relation may specify a differentiable branch without solving explicitly for it. A parameter may specify both coordinates of a curve. We compute derivatives in these forms, then connect local derivatives to changes across an interval. The mean value theorem gives that connection without assuming continuity of the derivative.*

By the end of this lecture you will be able to:

- Differentiate a given differentiable implicit branch while checking division conditions.
- Calculate graph derivatives for twice differentiable coordinates when $x(t)$ is strictly monotone and $x'(t) \neq 0$.
- Prove Fermat's, Rolle's and Lagrange's theorems with their endpoint hypotheses.
- Infer monotonicity, constancy and error bounds from derivatives on an interval.

9.1 Differentiating a given branch

An equation involving x and y need not give a single value of y for each x. A *branch* on an interval I is a function $y : I \to \mathbb{R}$ whose values satisfy the equation at every $x \in I$. *Implicit differentiation* differentiates that identity after a differentiable branch has been given. The chain rule then includes a factor $y'(x)$ whenever an expression depends on $y(x)$. This procedure does not prove existence or differentiability of a branch. A later several-variable theorem in Lecture 37 will address local existence.

For example, suppose a differentiable branch satisfies

$$x^2 + xy(x) + y(x)^2 = 3.$$

Figure 9.1 shows the two root choices for this relation. The product and chain rules give

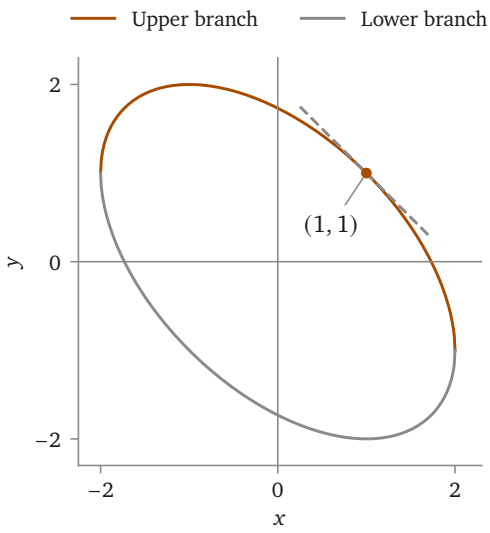


Figure 9.1: The two branches of $x^2 + xy + y^2 = 3$ for $-2 \le x \le 2$. The dashed tangent at $(1, 1)$ has slope -1.

$$2x + y(x) + xy'(x) + 2y(x)y'(x) = 0.$$

If $x + 2y(x) \neq 0$, collecting the terms gives

$$y'(x) = -\frac{2x + y(x)}{x + 2y(x)}. \tag{9.1}$$

At the point $(1, 1)$ this gives derivative -1. There is an actual branch there: the positive-root choice

$$y(x) = \frac{-x + \sqrt{12 - 3x^2}}{2}$$

is defined and differentiable on $(-2, 2)$ and satisfies the relation. Substitution into the quadratic in y verifies the identity, and $y(1) = 1$. Its derivative from the explicit formula at 1 is also -1. The other root choice is another branch, not another value of this function.

The nonzero coefficient in Eq. (9.1) is essential. For the relation $y^2 = x^2$, both $y = x$ and $y = -x$ are differentiable branches through $(0, 0)$. Differentiating gives $2yy' = 2x$. At the origin this becomes $0 = 0$ and determines neither derivative. The two actual branch derivatives there are 1 and -1. An invalid division by $y = 0$ would discard this distinction.

9.2 Higher derivatives and parametric graphs

If the derivative function f' is itself differentiable, its derivative is the *second derivative*, written f''. Repeated differentiation defines $f^{(n)}$ for integers $n \ge 1$ wherever each preceding derivative is defined and differentiable. The notation $f^{(0)} = f$ denotes the original function. For an open interval I, the class $C^n(I)$ consists of functions with derivatives through order n and continuous nth derivative on I. Continuity of the earlier derivatives follows from their differentiability.

For position $s(t)$ along a line, the first derivative is velocity and $s''(t)$ is *acceleration*, the rate of change of velocity. If $s(t) = t^4 - 2t^3$ with position in metres and time in seconds, then

$$s'(t) = 4t^3 - 6t^2, \qquad s''(t) = 12t^2 - 12t.$$

At $t = 2$, the velocity is $8\,\mathrm{m/s}$ and acceleration is $24\,\mathrm{m/s^2}$. Their units and physical roles differ.

For the explicit branch above, the square root has positive argument near 1 and its elementary derivatives can be taken twice. Differentiating the first implicit identity again gives

$$2 + 2y'(x) + 2[y'(x)]^2 + (x + 2y(x))y''(x) = 0.$$

At $x = 1$, with $y(1) = 1$ and $y'(1) = -1$, this becomes $2 + 3y''(1) = 0$. Thus $y''(1) = -2/3$. The same nonzero coefficient justifies solving for the higher derivative.

A *parametrised plane curve* assigns coordinates $x(t)$ and $y(t)$ to each real parameter t in an interval J. To express it as a graph, assume both coordinates are differentiable on an open interval J, with x strictly monotone and $x'(t) \neq 0$ there. Differentiability also makes x continuous. Its differentiable inverse gives $t = t(x)$, so the graph is $Y(x) = y(t(x))$. The chain and inverse rules give

$$\left.\frac{dY}{dx}\right|_{x=x(t)} = \frac{y'(t)}{x'(t)}. \tag{9.2}$$

If x and y are twice differentiable, applying the same rule to this quotient gives

$$\left.\frac{d^2Y}{dx^2}\right|_{x=x(t)} = \frac{1}{x'(t)}\frac{d}{dt}\left(\frac{y'(t)}{x'(t)}\right) = \frac{y''(t)x'(t) - y'(t)x''(t)}{[x'(t)]^3}. \tag{9.3}$$

Thus a second graph derivative is not generally $y''(t)/x''(t)$. Each differentiation with respect to x uses the inverse parameter again.

Figure 9.2 shows the positive-parameter graph used below.

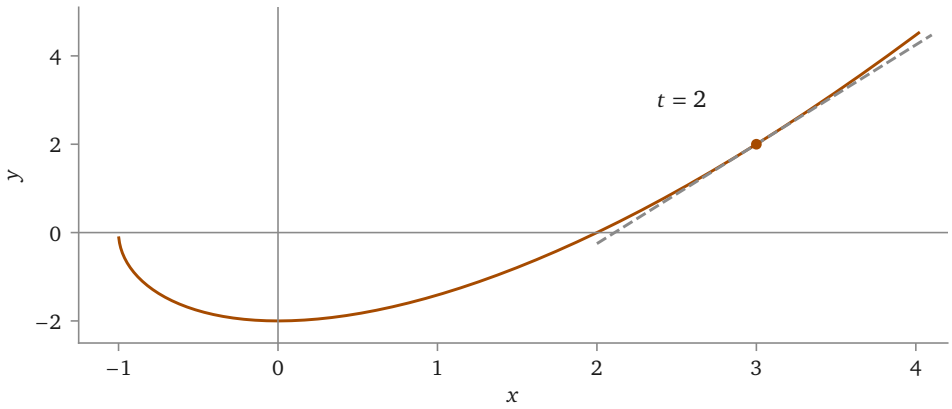


Figure 9.2: The curve $x = t^2 - 1$, $y = t^3 - 3t$ for positive t. The dashed line is its graph tangent at the parameter value $t = 2$.

For example, take $x(t) = t^2 - 1$ and $y(t) = t^3 - 3t$ for $t > 0$. The first coordinate is strictly increasing and has derivative $2t > 0$, so the graph interpretation is valid on this parameter interval. The formulas give

$$\frac{dY}{dx} = \frac{3t^2 - 3}{2t}, \qquad \frac{d^2Y}{dx^2} = \frac{3(t^2 + 1)}{4t^3}.$$

At $t = 2$, the curve point is $(3, 2)$, its graph slope is $9/4$, and its second graph derivative is $15/32$. The tangent line is therefore $y = 2 + (9/4)(x - 3)$. At $t = 0$ the first-coordinate derivative vanishes, so these graph formulas cannot be used there. No value is assigned by cancelling a zero denominator.

9.3 Interior extrema and Rolle's theorem

A function $f : A \to \mathbb{R}$ has a *local maximum* at $c \in A$ if some $\delta > 0$ gives $f(x) \leq f(c)$ for all $x \in A$ with $|x - c| < \delta$. A *local minimum* reverses the inequality. These are local extrema. Unlike a global extremum, the comparison is only with inputs in one neighbourhood.

Theorem 9.1 (Fermat's theorem). *If f has a local extremum at an interior point c of its interval domain and is differentiable at c, then $f'(c) = 0$.*

Proof. At a local maximum, sufficiently small positive increments satisfy $[f(c+h) - f(c)]/h \leq 0$. The same numerator for negative increments gives a quotient at least zero. Both one-sided quotient limits equal the finite derivative. Passing the inequalities to these limits gives $f'(c) \leq 0$ and $f'(c) \geq 0$, hence $f'(c) = 0$. For a local minimum, apply this maximum argument to $-f$. Its derivative is $-f'(c)$, so the conclusion is unchanged. ∎

The hypotheses distinguish three different cases. The function x^2 on $[-1, 1]$ has an interior minimum at zero and its derivative is zero there. The function $|x|$ also has an interior minimum there, but has no derivative. The identity on $[0, 1]$ has its minimum at the endpoint zero, with right derivative 1. Fermat's theorem does not assert a zero derivative in the last two cases. A zero derivative is not sufficient for an extremum: x^3 has derivative zero at zero and takes negative values on one side and positive values on the other.

Theorem 9.2 (Rolle's theorem). *Let $a < b$. If $f : [a, b] \to \mathbb{R}$ is continuous, differentiable on (a, b), and $f(a) = f(b)$, then some $c \in (a, b)$ satisfies $f'(c) = 0$.*

Proof. If f is constant, every interior point has derivative zero. Otherwise the extreme value theorem gives a maximum and a minimum, and at least one of these values differs from the common endpoint value. An input attaining that different value cannot be either endpoint. It is an interior global, hence local, extremum. Fermat's theorem gives zero derivative there. ∎

For example, $f(x) = x^2 - 3x$ on $[0, 3]$ has equal endpoint values zero. It is continuous and differentiable, so Rolle's theorem applies. Its derivative is $2x - 3$, which vanishes at the interior point $c = 3/2$. In contrast, $|x|$ on $[-1, 1]$ has equal endpoint values but no point with derivative zero. Its failure of differentiability at zero prevents application of Rolle's theorem.

9.4 Lagrange's mean value theorem

Theorem 9.3 (Lagrange's mean value theorem). *Let $a < b$. If $f : [a, b] \to \mathbb{R}$ is continuous and differentiable on (a, b), then some $c \in (a, b)$ satisfies*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Set $m = (f(b) - f(a))/(b - a)$ and define $g(x) = f(x) - mx$. The endpoint difference is $g(b) - g(a) = f(b) - f(a) - m(b - a) = 0$. Thus g satisfies Rolle's theorem on $[a, b]$. At its supplied interior point c, the derivative is $g'(c) = f'(c) - m = 0$. Rearranging proves the formula. ∎

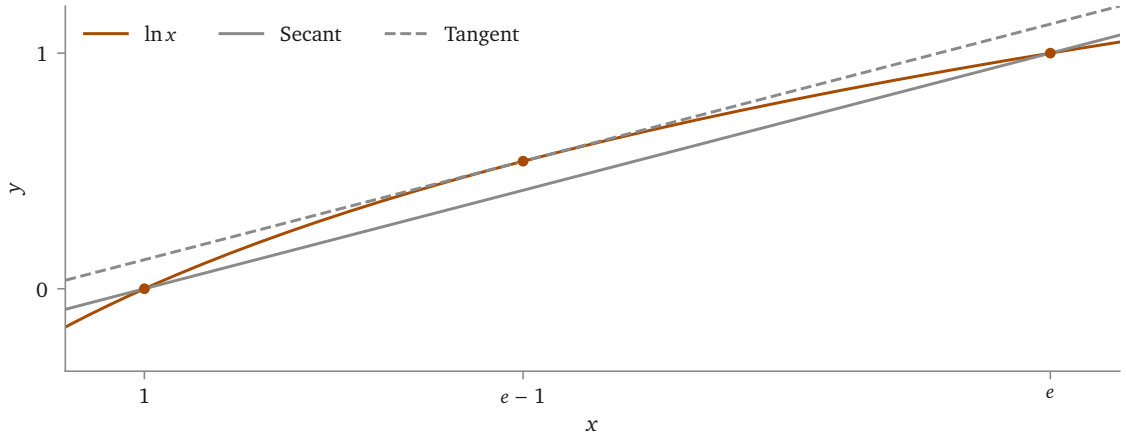


Figure 9.3: The logarithm on $[1, e]$. The endpoint secant and the tangent at $c = e - 1$ are parallel, with the same slope $1/(e-1)$.

The mean value theorem equates the slope of the endpoint secant to a tangent slope at an interior input. Figure 9.3 illustrates this equality for the logarithm.

For $f(x) = \ln x$ on $[1, e]$, the secant slope is $1/(e-1)$. Solving $f'(c) = 1/c = 1/(e-1)$ gives $c = e - 1$. The bounds $2 < e < 3$ show $1 < c < e$, as the theorem requires. The tangent has equation $y = \ln(e-1) + (x - (e-1))/(e-1)$. The theorem supplies existence for general functions. In this example the derivative formula also identifies the point explicitly.

For a motion $s(t)$ continuous between times a and b and differentiable at all intermediate times, the theorem says that its average velocity $[s(b) - s(a)]/(b-a)$ equals its instantaneous velocity at some intermediate time. It does not identify that time or assert that it is unique. Continuity of velocity is not among the theorem's hypotheses.

9.5 Monotonicity, constancy and bounds

Recall that $f : I \to \mathbb{R}$ is nondecreasing if $f(u) \le f(v)$ whenever $u < v$ are in I. It is nonincreasing if the output inequality is reversed. These definitions permit equal outputs at distinct inputs.

Theorem 9.4 (Derivative signs on an interval)**.** *Let f be continuous on a real interval I and differentiable at every interior point. If $f' \ge 0$ throughout the interior, then f is nondecreasing on I. If $f' > 0$ there, then f is strictly increasing. The corresponding statements with nonpositive or negative derivatives give nonincreasing or strictly decreasing functions. If $f' = 0$ throughout the interior, then f is constant on I.*

Proof. Choose any $u < v$ in I. The closed interval $[u, v]$ is contained in I. Its interior points are interior points of I, so the mean value theorem applies and gives

$$f(v) - f(u) = f'(c)(v - u) \qquad \text{for some } c \in (u, v).$$

Since $v - u > 0$, a nonnegative derivative makes the difference nonnegative. A positive derivative makes it positive. Applying these conclusions to $-f$ gives the two decreasing statements. If the derivative is zero throughout, the displayed difference is zero for every pair, proving constancy. On an interval with at most one point, the same conclusions hold directly. ∎

Conversely, a differentiable nondecreasing function has nonnegative derivative: for positive increments the numerator and denominator of the quotient are nonnegative, and for negative increments both are nonpositive. Every quotient is nonnegative, so its limit is nonnegative. Strict increase does not force a strictly positive derivative everywhere, as x^3 at zero demonstrates. The interval hypothesis for constancy also matters. A function equal to zero on $(-\infty, 0)$ and 1 on $(0, \infty)$ has derivative zero at every point of its domain, but is not constant on that disconnected domain.

The same proof yields quantitative bounds. If $|f'(x)| \le M$ at every interior point of I, where $M \ge 0$, then

$$|f(v) - f(u)| \le M|v - u| \qquad (u, v \in I). \tag{9.4}$$

For distinct inputs this follows by taking absolute values of the mean value identity, and equal inputs give zero on both sides. For example, $f(x) = \arctan x$ has $0 < f'(x) \le 1$ on $\mathbb{R}$. It is therefore strictly increasing and $|\arctan v - \arctan u| \le |v - u|$ for all real u and v. An input error at most $1/100$ changes this angle by at most $1/100$ radian. This is a finite error guarantee, unlike an approximation that only controls a remainder in the limit.

The logarithm gives a further exact inequality. For $x > 0$, apply the mean value theorem to $\ln t$ on $[1, 1 + x]$. Some $c \in (1, 1 + x)$ gives $\ln(1 + x) = x/c < x$. For $-1 < x < 0$, use $[1 + x, 1]$ instead. Some $c \in (1 + x, 1)$ again gives $\ln(1 + x) = x/c < x$, since $x < 0$ and $c < 1$. Including $x = 0$ gives

$$\ln(1 + x) \le x \qquad (x > -1), \tag{9.5}$$

with equality exactly at zero. Its proof uses the logarithmic derivative, which was itself established from standard limits, so no inequality was assumed in deriving that derivative.

Summary 9.

- Implicit differentiation computes derivatives of a given differentiable branch. Solving for its derivative requires a nonzero coefficient. It does not establish existence of the branch.
- Equations (9.2) and (9.3): an invertible coordinate parameter with $x' \neq 0$ gives graph derivatives y'/x' and $(y''x' - y'x'')/(x')^3$. The first requires differentiable coordinates, and the second requires twice differentiable coordinates.
- Theorems 9.1 and 9.2: an interior differentiable extremum has derivative zero. Equal endpoint values and Rolle's other hypotheses guarantee such a zero derivative somewhere inside.
- Theorem 9.3: continuity on $[a, b]$ and differentiability on (a, b) give an interior derivative equal to the secant slope.
- Theorem 9.4 and Eq. (9.4): derivative signs control monotonicity on intervals. Zero derivatives imply constancy, and bounded derivatives give finite bounds on output changes.

Lecture 10 proves Cauchy's mean value theorem and uses it to resolve indeterminate quotients. It then studies how derivative signs describe a graph. Sheet 9 practises implicit and parametric derivatives and mean value arguments.

References

R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, Chapter 6.